HemOnc FHIR Regimen Catalog

Executable Knowledge Layer for Oncology Treatment Protocols

Beyond OMOP: what HemOnc's additional tables add — with worked examples for FOLFOX4 and a chemo + targeted kinase inhibitor combination

Draft / experimental — 2026-05-27

Developed in cooperation with HemOnc.org · OHDSI Oncology · HL7 Europe PHOENIX · MII Onkologie Working Group

Not a medical device. Not for clinical use without independent validation.

1. Two layers of HemOnc, only one of them widely known

The HemOnc Knowledge Base (HemOncKB) is the canonical oncology vocabulary in the OMOP Common Data Model, distributed via Athena/OHDSI since 2019 (Warner JL et al., *J Biomed Inform*). What most consumers see is the OMOP-format core: three flat triple-shaped tables that list concepts, relationships and synonyms. Useful, but intentionally limited — OMOP triples cannot represent dosing schedules, target-AUC computations, multi-component protocols, or evidence-graph relationships. Those live in the eighteen **additional tables** that ship alongside the OMOP core and that, until now, have not had a structured machine-readable consumer interface. This IG is that consumer interface.

Two layers, two consumers

Layer	Tables	Rows (snapshot 2026-03-12)	Typical consumer
OMOP core	concept_stage, concept_relationship_stage, concept_synonym_stage	~800 k	OHDSI research sites, observational analytics
Additional tables	18 tables across 6 functional groups	~310 k	This IG — clinical-ordering, CDS, evidence linkage

2. What the additional tables carry, organized by purpose

The 18 additional tables group into six functional clusters. Together they carry the content that makes HemOnc clinically actionable — not just terminology, but executable protocols with full provenance.

Executable protocol content — *the heart of the catalogue*

Table	Rows	Adds
regimens	2 794	Regimen-family metadata: type, modality, component count, study count, highest evidence, first studied year, last published date
variants	27 393	Per-(study, regimen, variant) full encoding of the dosing protocol. Self-contained — no diffs, no inheritance
sigs	24 555	Atomic dosing unit: drug, dose (mg/m², AUC, mg/kg), route, day pattern, frequency, cycle length, infusion duration, role, relative sequencing
variant_eligibility	635	Dose-eligibility criteria: age, weight, lab thresholds

Drug knowledge — *beyond a code list*

Table	Rows	Adds
drugs	873	INN, ATC, main_class (Platinum agent, Taxane, Anti-PD-1, …), CanMED axis, investigational + multi-agent flags
rxnorm_map	901	HemOnc ↔ RxNorm crosswalk with semantic type (IN, SCD, …)
indications	3 549	FDA/EMA approvals by condition: date, accelerated, withdrawn, biomarker, prior-therapy, demographics, narrative reason

Disease conditions — *with eight crosswalks*

Table	Rows	Adds
conditions	318	HemOnc disease concepts with mappings to NCIt, SNOMED CT, OncoTree, ICD-9-CM, ICD-10-CM, ICD-O-3, ICD-O-3 morphology, SEER

Evidence base — *the EBM backbone*

Table	Rows	Adds
studies	7 697	Clinical trials: registry (CT.gov, EudraCT, …), trial ID, condition, biomarker, enrolment, phase, arms, design, sponsor, group
study_results	14 727	RCT endpoints with HR/OR/RR, CI bounds, p-values, comparator code, landmark — maps to FHIR EffectEvidenceSynthesis
refs	11 259	Publications: PMID, DOI, PMCID, title, journal, biblio, pub-date, ref type, errata, citation count
authors	180 345	Per-publication author records with order, role, affiliation
persons	53 545	Per-author rollup, normalized across publications

Curation transparency — *the epistemic honesty layer*

Table	Rows	Adds
inclusions	10 494	Publications HemOnc accepted with the reason (criteria-driven or SOP-exception)
exclusions	18 198	Publications HemOnc reviewed and rejected with the reason — bigger than inclusions, by design

Wiki linkage and schema-meta — *supporting structure*

Table	Rows	Adds
pointers	21 935	Versioned URL backlinks to the HemOnc.org wiki Table-of-Contents (narrative explanations, dose-modification rules)
hemonc_classes	148	Concept class definitions
hemonc_rels	95	Relationship type definitions
canonical.triples	220	Allowed (class■ — relationship — class■) combinations — the schema constraint

3. Worked example — FOLFOX4

FOLFOX4 is the canonical first-line systemic therapy for metastatic colorectal cancer and the prototype of bi-weekly cytotoxic combinations. It is in the MII Onkologie protocol code list as `FOLFOX4` and in HemOnc as regimen CUI 13829 with 5 variants across 28 studies. First studied 1995, last published 2024-11-15. *Phase 3 RCT* evidence level.

3.1 Regimen-level metadata

Field	Value
regimen_name	FOLFOX4
regimen_cui	13829
regimen_type	Component-based regimen
MII protocol code	`FOLFOX4` from `mii-cs-onko-systemische-therapie-protokolle` v2026.0.3
sact	TRUE (systemic anticancer therapy)
components	3 — Fluorouracil + Leucovorin + Oxaliplatin
studies	28
variants	5
highest_evidence	Phase 3 RCT
first_studied	1995
last_published	2024-11-15

3.2 Per-component sigs (canonical variant 132249, q14d cycle)

Component	Dose	Route	Day	Frequency	Duration	Step
Fluorouracil	400 mg/m²	IV	1, 2	once per day	1 min	1 of 2 (bolus)
Fluorouracil	600 mg/m²	IV	1, 2	continuous	22 h	2 of 2 (CI)
Leucovorin	200 mg/m²	IV	1, 2	once per day	2 h	1 of 1

Component	Dose	Route	Day	Frequency	Duration	Step
Oxaliplatin	85 mg/m²	IV	1	once	—	1 of 1

Cycle length 14 days. The four sig rows above correspond to four FHIR `ActivityDefinition` resources, orchestrated by one `PlanDefinition` for the variant. Each ActivityDefinition is `kind = MedicationRequest`, references a `MedicationKnowledge` by canonical, and carries the dose as `doseQuantity { value, unit: 'mg/m2', system: UCUM }`. At `$apply` time the Library's BSA helper multiplies by the patient's BSA to produce the absolute mg dose.

3.3 Regulatory indications

Regulator	Date	Condition	Stage	Prior therapy
FDA	2002-08-09	Colorectal cancer	Metastatic	after FOLFIRI
FDA	2004-01-09	Colorectal cancer	(any)	—

Each row becomes a `PlanDefinition.useContext` entry plus a `relatedArtifact[type=documentation]` pointing at the regulatory label. Biomarker-conditional approvals (KRAS-WT, BRAF V600E) also live in this table — they generate `PlanDefinition.action.condition` CQL expressions at the catalog level so downstream CDS Hooks responses can be biomarker-aware without the calling EHR encoding the rule itself.

3.4 Supporting studies (selected, by appearance frequency)

Study	Trial ID	Condition	Phase	First publication
MOSAIC	—	Colorectal cancer (adjuvant)	Phase 3	André T et al., NEJM 2004
NCCTG N9741	—	Metastatic colorectal cancer	Phase 3	Goldberg RM et al., JCO 2004
EORTC 40983 (EPOC)	NCT00006479	CRC liver metastases	Phase 3	Nordlinger B et al., Lancet 2008
PRIME	NCT00364013	Metastatic CRC (KRAS-WT)	Phase 3	Douillard JY et al., NEJM 2013
EACH	NCT00471965	Hepatocellular carcinoma	Phase 3	Qin S et al., JCO 2013
IDEA	—	Colon cancer (adjuvant)	Phase 3	Grothey A et al., NEJM 2018

Every study maps to a FHIR `ResearchStudy` with the registry ID as `identifier`. Every published endpoint result maps to `EffectEvidenceSynthesis` (HR, CI, p). The PlanDefinition carries `relatedArtifact[type=derived-from]` pointing at each supporting study and at the primary publication via DOI/PMID. **This is the closing-the-loop instrumentation** — every administered dose at a downstream site traces back, via the knowledge artifact, to the trial that justified it.

4. Worked example — chemo + targeted kinase inhibitor combination

mFOLFOX6 + Lenvatinib + Pembrolizumab (HemOnc cui 28874) is a contemporary five-component regimen combining a classical cytotoxic backbone with a multikinase inhibitor (Lenvatinib, ATC L01EX08, hits VEGFR-1/2/3, FGFR-1/2/3/4, RET, KIT, PDGFRα) and a checkpoint inhibitor (Pembrolizumab, anti-PD-1 mAb). Phase 3 RCT evidence, first studied 2021, five HemOnc components, three sact modalities (cytotoxic, targeted, immune). Demonstrates that the same FHIR catalog mechanics scale from a 1995-era cytotoxic protocol to a 2021-era multi-modality regimen with no structural changes to the IG.

4.1 The six sigs, side-by-side

Component	Class	Dose	Route	Day pattern	Cycle
Fluorouracil	Fluoropyrimidine	400 mg/m²	IV bolus	day 1	14 d
Fluorouracil	Fluoropyrimidine	2 400 mg/m²	IV CI	day 1	14 d
Leucovorin	Folate-reduction agent	400 mg/m²	IV	day 1	14 d
Oxaliplatin	Platinum agent	85 mg/m²	IV	day 1	14 d
Lenvatinib	Multikinase inhibitor (TKI)	8 mg	Oral	continuous	42 d
Pembrolizumab	Anti-PD-1 antibody	400 mg	IV	day 1	42 d

Two cycle lengths in one regimen (q14d chemo + q42d Lenvatinib/Pembro) is a common modern pattern that the catalog handles natively via nested `PlanDefinition.action` groups with independent `timingTiming`. The dose-target paradigm differs by component: mg/m² for cytotoxics (multiply by BSA), absolute-mg for Lenvatinib (no patient scaling) and Pembrolizumab (the FDA-approved fixed dose), with provenance preserved per-component via the `sig-target-type` CodeSystem.

4.2 Dose-modification surface for Lenvatinib

Lenvatinib has the most clinically active dose-modification rule of any drug in this regimen: starting 8 mg PO daily continuous, reduced to 4 mg every other day after grade 3 adverse events, with discontinuation thresholds. None of this lives in HemOnc's structured tables — it is in the FDA label and in the HemOnc.org wiki narrative reached via the `pointers` backlink. In our IG architecture, dose-modification rules become a separate institution-authored CQL Library that the catalog PlanDefinition references via `relatedArtifact[type=depends-on]`. Site policy plugs in at this seam without touching the catalog content. *This is why we deliberately separate protocol-knowledge from institution-policy from order-banding from administration-recording.*

5. Why this matters — closing the loop

Today, oncology protocols live as PDFs in pharmacy binders, as proprietary order sets inside individual EHR vendors (NCCN Templates™, Epic Beacon, Cerner PowerPlans), and as free-text Protokoll fields in cancer registry submissions. There is no openly-licensed, executable, versioned, machine-actionable catalogue of oncology regimens at European or US scale. HemOncKB has the content; OMOP has the vocabulary; this IG supplies the missing piece — a FHIR R4 projection of HemOnc that downstream EHR / CDS / registry / research systems can address by canonical URL.

The strategic value is not the catalog itself. The strategic value is the round-trip: an institution administering a HemOnc-derived FOLFOX4 cycle today, with structured MedicationRequests carrying provenance back to the trial that justified the dose, produces structured MedicationAdministration data tomorrow that flows back through MII Onkologie / mCODE / PHOENIX ECCM into OMOP, where the next HemOnc curation cycle sees the real-world outcome. **Knowledge** → **care delivery** → **real-world evidence** → **knowledge.** One vocabulary on both sides of the loop, instead of two.

5.1 What is in the repository today

Artifact	Status	Notes
sushi-config.yaml + ig.ini	scaffold ready	IG-Publisher consumable; declares dependency on de.medizininformatikinitiative.kerndatensatz.onkologie 2026.0.3
8 hand-authored R4 JSON resources	validated	Carboplatin + Paclitaxel Variant #83 end-to-end (2 CodeSystem, 2 MedicationKnowledge, 2 ActivityDefinition, 1 Library, 1 PlanDefinition)
CQL Library + FHIRPath dynamicValues	validated	BSA + Calvert + provenance codes; cqf-fhir 8.4.0 quirks documented in beads memory
HAPI FHIR + cqf-fhir Docker compose	running locally	Port 8096; load + $apply round-trip green; demo Patient + Observations produce Carbo 625 mg + Taxol 323.75 mg
7 IG narrative pages	first draft	index, background, cooperation, disclaimers, versioning, $apply/CDS, phase 0
beads issue tracker	13 open issues	Phase 1A scope, MII∩HemOnc match script, PHOENIX engagement, regulatory framing, IP posture, ConceptMap, etc.

5.2 What this is not

This is not a medical device. It is reference content under MDR/IVDR and 21 CFR. It is not CE-marked, FDA-cleared, or otherwise regulatorily conformed. Integration into any system intended to influence individual patient care creates an obligation on the integrator — not on the publisher — to perform appropriate regulatory qualification and conformity assessment of that downstream system. See the IG's *Disclaimers and Intended Use* page for the full statement.

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